

Modelling duel-pheromone ant foraging in a multi-node environment

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April 2025

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Abstract

This paper presents a unique duel-pheromone solution for an ant foraging model in a multi-node environment. This alternative method addresses the limitations in scalability that can arise in previous traditional approaches, where a new unique pheromone type is assigned to each food node, leading to a linear increase in computational complexity for every new node added to the environment. The method outlined in this paper simplifies these traditional methods by making use of a constant two unique pheromone types for any number of food nodes through the use of popularity-based trail following selection, providing a more straightforward decision making process for agents, as well as a more constant computational complexity. This model also allows for not only nest to food node pheromone trails but food to food connections too, by treating a newly discovered food node as another nest node. Similar to previous work the environment is made up of a grid world, where each cell provides both environmental objects and pheromone concentrations, all encoded into one integer, with both evaporation and diffusion mechanics included to imitate natural pheromone dynamics. The resulting model is capable of forming low cost, low robustness network structures, connecting all nodes in a small scale environment, and the majority of nodes on a larger scale. These experiments demonstrate how the model can reliably generate optimized pheromone networks for varying configurations, outperforming earlier models in scalability and computational efficiency. The model shows promising applications in the real world, providing potential solutions in areas such as city transport route optimizations and swarm robotics. Future work on the model will explore more complex environments containing obstructions and moving food nodes, as well as adaptive parameter tuning.

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1 Introduction

Social insects, such as ants, bees, wasps, and termites, are among the most sought after models for multi-agent research, as they yield an advanced collective intelligence from large colony populations while each following a relatively simple set of instructions. The swarm behaviours that these insects showcase have led to inspirations in research for swarm robotics^[1–5], including navigation, path planning detection systems, negotiation protocols, and cooperative control^[3]. Furthermore, studies have shown that navigation and pathing behaviours extend outside of social insects, and are found in systems such as vertebrate vascular systems^[6], and mycelial networks of fungi^[7]. Navigation and pathing algorithms are a core construct for many species, including humans, and many network routing algorithms are inspired by social insects, making use of a population-oriented stochastic search methods^[8].

Popular methods to achieve swarm behaviour models rely on *stigmergy*, where individuals in a group communicate indirectly through the environment. An example of this is a pheromone based communication procedure between agents, a chemical substance that is deposited by ants, as well as similar social insects (Dorigo et al.^[4]). These pheromone deposits indirectly relay information to other ants within the colony, notably to mark the environment with a path to and from a food source or other

points of interest. Dorigo et al., as well as many others, showcase how these social behaviours can be used and manipulated to solve a wide range of problems, including, but not limited to, the Travelling Salesman, traversing an environment with obstacles^[5,9], and the Job Shop Scheduling and Assignment problem^[10]. Further studies have investigated how varying species of ants will form differently shaped pheromone trail networks, optimizing for its efficiency, robustness, and cost depending on the species' needs^[11,12].

Panait and Luke^[9] built an agent based ant model for traversing an object filled environment, with the agents able to rapidly find, reinforce, and optimize a pheromone trail between *nodes*, representing nest and food sources. Their model made use of multiple pheromones, with every additional node requiring its own unique pheromone type. This model works well for a small number of total nodes within the environment, leading to the ants forming an efficient and robust pheromone trail network. However, for every node added to the environment, the computational intensity linearly increases, due to the increased amount of pheromones within the environment needing to be updated, causing a bottleneck when simulating an environment with a great number of nodes.

Although Panait and Luke's model does not cause computational issues for most study simulations where the environments node count is low, it does present issues when simulating on larger scales for purposes such as generating optimal network connections for city transport, where a large quantity of nodes are required. The model in this paper proposes a solution to this bottleneck, making use of only two pheromone types for any number of nodes. In doing this, the time complexity scale of pheromone updates reduces from linear to constant, and more closely relates to ant species such as big headed ants (*Pheidole megacephala*)^[13] and Pharaoh's ants (*Monomorium pharaonis*)^[14]. This also leads to a simpler model that holds fewer required agent states, as a decision for which pheromone to follow and lay next is not required, and instead an agent switches to the opposing state upon the arrival of a node.

Alternative studies propose models that are capable of forming pheromone trails to multiple food sources from the home nest, only utilizing either one^[15] (relying on an ad-hoc solution for returning home, such as an internal compass) or two^[16] pheromone types. However, these models are only able to maintain multiple pheromone trails to differing food sources if they are both equidistant from the nest, and will otherwise drop the furthest source.

The model presented in this paper is capable of maintaining pheromone trails between multiple food nodes, as long as they are within the agents' range, even if their distances from the nest node differ. Additionally to this, agents will expand their colonies range of exploration upon the discovery of food nodes, allowing for the creation of pheromone trails from one food node to another. This is achieved through a popularity rating of trails, where an agent will ignore a particular trail if it's popularity exceeds a certain threshold, differing from most other alternative models where agents follow the strongest pheromone at all times, or rely on random exploration for exploring addi-

tional routes.

This study begins with introducing the new model, explaining it's development and implementation. Applications of the model through simulated environments are then presented, showcasing the model's ability to forage, and create a network connecting all nodes. The paper comes to an end with discussing the results, as well as future workings and the different applications this model may have.

2 Professional Considerations and Ethics

2.1 BCS Code of Conduct

The relevant sections to this project of the BCS Code of Conduct are as follows:

2.1.1 Public Interest

(a) You shall have due regard for public health, privacy, security and wellbeing of others and the environment.

This simulation does not have any form of effect on the health, privacy, security or wellbeing of others, nor the environment. No data is held about users, and the health and wellbeing of others and the environment are unaffected as the simulations are not physically applied to the real world in this project.

(b) You shall have due regard for the legitimate rights of Third Parties.

This simulation makes no use of third party content. All assets and code will be self-designed and created. In the event that third party content is used, the content will be free to use, all license obligations will be met, and the creator will be properly credited.

(c) You shall conduct your professional activities without discrimination on the grounds of sex, sexual orientation, marital status, nationality, colour, race, ethnic origin, religion, age or disability, or of any other condition or requirement.

The simulation is unable to promote or exercise discrimination of any of the listed traits, as the simulation of ants has no relevance to them. Possible concerns could be raised about the sex or race of ants, which is of no concern as all ants will be equal with no specified sex nor race. The only differences between ants will be their ability to fulfil the task at hand.

(d) You shall promote equal access to the benefits of IT and seek to promote the inclusion of all sectors in society wherever opportunities arise.

This project has worked to accommodate users with colour blindness disabilities, using bright colours (blue and green) on a white background to easily distinguish visual information. Disabilities of hearing impairment are not relevant,

as the simulation has no sound, and similarly with motor disabilities, there are no simulation interaction features.

2.1.2 Professional Competence and Integrity

- (a) You shall only undertake to do work or provide a service that is within your professional competence.
 (b) You shall **NOT** claim any level of competence that you do not possess.
 (c) You shall develop your professional knowledge, skills and competence on a continuing basis, maintaining awareness of technological developments, procedures, and standards that are relevant to your field.

In accordance with these three points, 2(a,b,c), myself and my technical supervisor are both confident in my ability to achieve the outlined objectives, and that the project as a whole is within my professional competence.

- (d) You shall ensure that you have the knowledge and understanding of Legislation and that you comply with such Legislation, in carrying out your professional responsibilities.

I have been sure to not break any legislative laws throughout the duration of this project's lifetime.

- (e) You shall respect and value alternative viewpoints and, seek, accept and offer honest criticisms of work.

While working on the project, I provided progress updates with my technical supervisor, as well as others on my course. Any feedback and/or criticism was acknowledged and presented without edit or omission.

- (g) You shall reject and will not make any offer of bribery or unethical inducement.

I neither accepted nor promoted any form of bribery or unethical influence, and all feedback was collected without incentives tied to its sentiment.

3 Method

The model proposed in this study generally relies on Panait and Luke's model^[9] as a baseline reference, with rules for how agents follow deposited pheromones being the majorly differing factor. The aim of this model is for the agents to be capable of connecting not only their nest node, N_{nest} , to a food node, $N_{food,i}$, but also to connect to multiple food nodes, as well as one food node to multiple other food nodes, while utilizing only two pheromone types, $P = \{p_A, p_B\}$. Agents must achieve the formation of pheromone trails that connect all nodes within the environment together, without losing necessary trails due to evaporation, such that paths can be traversed to get from any node, N_A , to any other node, N_B . This section details how the environment setup and the agent behaviours have been constructed to achieve these goals.



Figure 1: An example of a single cell's value within the environment, scaled down to a 32-bit representation. Each cell holds a single 64-bit long integer that is split into chunks, with each section assigned different meanings. Section 1, item type, determines whether there is a nest, food, an obstacle, or nothing at the cell. Sections 2 and 3, Pheromone A and B, contain the concentration values of the two types of pheromones.

3.1 Environment Setup

Ant foraging dynamics and trail formation present a complex set of challenges, yet the use of a simple grid world can effectively capture their essential behaviours, while forgoing unnecessary, minute details. The formation of a trail between two nodes can be affected by distance from one another, in addition to food density (given a food node) and its size^[17]. This model focuses only on the influence of distance, abandoning the latter considerations of food density and size, and assumes each food node provides an infinite source of food.

Every cell in the grid world contains one unsigned 64-bit long integer. Each integer contains three segments of information, as exemplified in Figure 1. The first two bits represent what object is at that particular cell, e.i., an obstacle that an ant cannot travel through, a food node for an ant to "pick up" food, a nest node to "deposit" food, or nothing. The next fifteen bits then hold the pheromone concentration value in the cell for pheromone p_A , with the following fifteen representing the p_B pheromone.

The behaviours of pheromones vary slightly across models, with either both diffusion and evaporation mechanisms being utilized, or only one of the two. In this study, the model relies on both diffusion (eq. 2) and evaporation (eq. 3) mechanics, with the two pheromone types behaving identically. This leads to greater simplicity of the model as a whole, eliminating the need for a complex sensor cone algorithm for each agent, where instead they can sense pheromones that are in their immediate vicinity with the same effect.

$$\bar{p}_{i,j} = \frac{1}{9} \sum_{x=-1}^1 \sum_{y=-1}^1 p_{(i+x,j+y)} \quad (1)$$

$$p'_{i,j} = p_{i,j} + D \cdot (\bar{p}_{i,j} - p_{i,j}) \quad (2)$$

In Equation 2, a given pheromone concentration of type $p \in P$, at grid point (i, j) updates by considering the average concentration of its immediate neighbours, $\bar{p}_{i,j}$, inclusive of itself (eq. 1). The diffusion factor, D , is a constant between 0 and 1, defining the rate at which pheromones P diffuse across the environment.

$$p'_{i,j} = p'_{i,j} \cdot (1 - k) \quad (3)$$

Equation 3 then applies evaporation to the resulting value of Equation 2, with k , another constant between 0 and 1,

denoting the evaporation factor. k represents what percentage of pheromone $p_{i,j}$ should be removed every simulation second.

In any given environment, it is assumed that there is one nest node where all agents begin, and some n food node locations for agents to forage to. Although theoretically multiple nest nodes may exist, as well as none at all, given that each agent is given an alternative position to start at. For simplicity, this section will start out assuming one nest node and one food node in examples.

3.2 Agent Behaviours

With an environment consisting of only one nest node and one food node, the problem at hand can be seen as the “central nest foraging” problem, where agents are to exit the nest in search for food, and return to the nest with the food they find. Upon leaving the nest, from any given grid cell, agents may move to any one of eight surrounding cells that does not contain an obstacle. When an agent comes across a food node, they become laden with food, where they then return to the nest to deposit the food and continue their foraging.

In an abstract view, the foraging task may be seen as a series of two interchanging steps for each individual agent. They start from the nest node to reach a food node, and then start at the food node to reach the nest node laden with food. An agent relies on its own personal internal state of whether they are laden with food or not, which determines the current task that they must be performing. The performance for each of these task state transitions can provide rewards upon arrival of the *goal state*: where the food node is the goal state when not laden with food, and the nest node is the start state, or vice versa.

As Panait and Luke^[9] explained, this approach is akin to reinforcement learning^[18], where each state is assigned a value that represents the cumulative reward. In this model, every grid cell (state) has a particular concentration value for both pheromone types of P . Each concentration is treated as a *utility value*, U_p , indicating the desirability of that state, given a pheromone type $p \in P$, in order to reach the goal state. Agents then utilize these values, $U_p(s)$, dependent on its current location (*external state*) s , to decide which state to move to next. When an agent then reaches their goal state (nest or food node, depending on its internal state), it receives a reward $R(s)$.

Taking Figure 2 as example, the agent has transitioned from state s to s' , and now has available state transitions $s'' \in S''$. The decision of which s'' to transition to is weighted by each utility value of S'' , with the focal pheromone, p , depending on the agent’s internal state and depicting which pheromone P to follow. If the agent is laden with food, p_{nest} will be the focal pheromone, else p_{food} if not laden with food and foraging instead (in this example of one nest node and one food node, $p_A \rightarrow p_{food}$ and $p_B \rightarrow p_{nest}$, for additional simplicity).

When transitioning from s' to some s'' (as well as $s \rightarrow s'$), the agent must deposit some amount of pheromones into s' , therefore updating $U_p(s')$, in order to create or reinforce a trail leading towards the agent’s previous goal

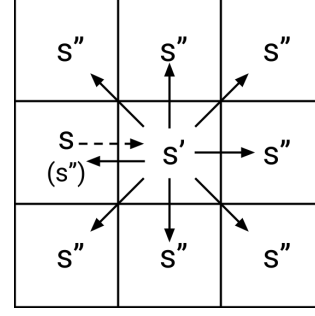


Figure 2: State transitions for any given agent, depicting all available states they can transition to, $s'' \in S''$, from their current state, s' , after transitioning from s . It is important to note that s is part of the set S'' too.

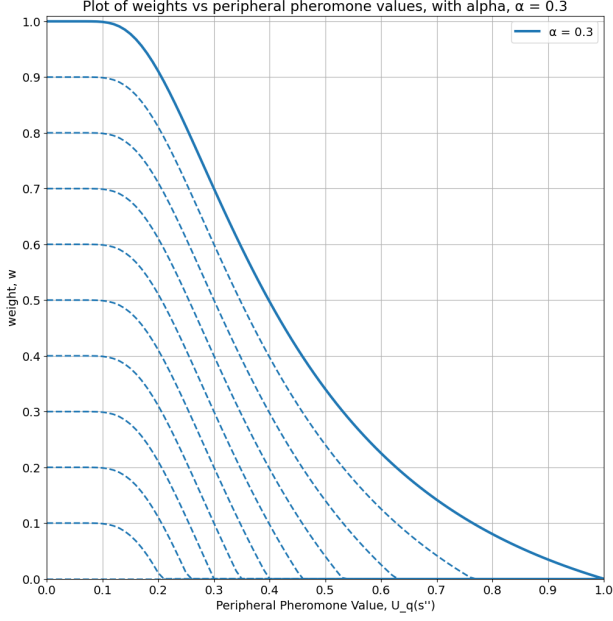
state. The pheromone type of p when depositing will be the opposite to that of the focal pheromone the agent is following, known as the peripheral pheromone (e.g., deposit p_{food} when laden with food and following p_{nest} back to the nest, or vice versa).

$$U_p(s') = R(s') + \gamma \max_{s'' \in S''} U_p(s'') \quad (4)$$

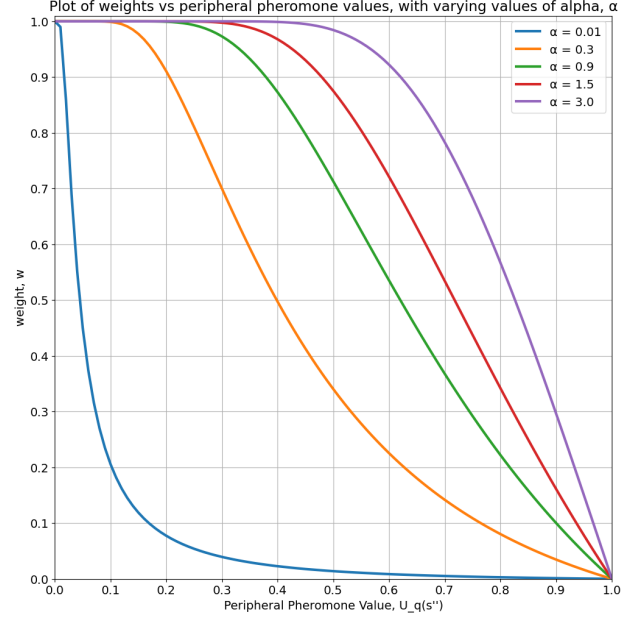
Equation 4^[9] describes the rule for updating $U_p(s')$, according to the reward of $R(s')$ and a decay rate γ . The value of γ is a constant value between 0 and 1, although Panait and Luke chose to make this value smaller for diagonal transitions, as to reflect the slightly longer distance. The model in this paper does not incorporate this varying value of γ .

The segment $\max_{s'' \in S''} U_p(s'')$ of Equation 4 selects the highest utility value of S'' , expanding upon the sequence of high concentration pheromones of type p that lead towards the agent’s start state node, or alternatively known as it’s previous goal state. The applied decay rate γ to this highest utility value gradually decreases the intensity of this pheromone trail the further away from the start state the agent gets, therefore making the trail less favourable the greater the distance that particular node is from the goal state. A reward $R(s')$ is then added onto this decayed utility value, in order to amplify the pheromone concentration when a desirable state is found. This reward is usually of value 0 when the agent is foraging or returning to the nest, but will be greater than 0 when at a goal state. This model has the reward of any goal state set to the maximum pheromone value possible, in order to reward all goals equally and greatly.

Due to this equation gradually decreasing the amount of pheromone deposited by a given agent, there is the guaranteed possibility that $U_p(s')$ becomes 0 if no goal state is found. If the agent were to continue on a foraging path while depositing zero pheromone, they would have no way of finding their way back to their start state upon arrival of a goal state, as their guiding trail back would no longer extend to their position. Due to this issue, agents will switch states with zero reward, $R(s')$, once the pheromone value that they are depositing becomes zero. This will cause them to turn around and head back to their start



(a) How the resulting weight, w , is effected by varying values of $U_p(s'')$, with $\alpha = 0.3$. Each line represents differing values of $U_p(s'')$ in the range of $[0, 1]$ with a step of 0.1. The bold line represents $U_p(s'') = 1$.



(b) How the resulting weight, w , is effected by varying values of α , with $U_p(s'') = 1$. Each line represents differing values of the popularity factor, $\alpha \in \{0.01, 0.3, 0.9, 1.5, 3.0\}$.

Figure 3: (a) and (b) graphically present how the focal pheromone utility value, $U_p(s'')$, is effected by the peripheral pheromone utility value, $U_q(s'')$, through Equation 5. It is important to note that the value of $U_p(s'')$ can be found for each line via looking at the w value where $U_q(s'') = 0$, as this means no change has been made to the focal pheromone's utility value, and thus $w = U_p(s'')$.

state, while not depositing any pheromone because they have not found anything. Upon arrival back to their start state, they will return to their foraging state and continue foraging. This process creates a exploration *range* from a given node, where agents cannot explore beyond a certain radius from the start state.

A similar processes is also applied with an internal timer for foraging, where an agent will return back to their start state if they have been foraging for too long (e.g. 60 simulation seconds). This prevents agents from indefinitely getting stuck in one spot, given that they have found a way to do so.

Once a pheromone has been deposited into s' , the agent must then select a state from S'' to transition into. In order to ensure all states in the environment are eventually explored, agents choose a uniformly random transition according to some probability. In all other cases where this probability is not met, the agent transitions into the state with the greatest *weight*, $w \in W$. The weights, W , for all *normalized* S'' values (in relation to the maximum possible utility value) are determined by Equation 5, where the state with greatest w is selected to transition into with Equation 6:

$$W = \{U_p(s'') - U_q(s'')^{\alpha/U_q(s'')} \mid s'' \in S''\} \quad (5)$$

$$s' = \underset{w \in W}{\operatorname{argmax}}(w) \quad (6)$$

This equation expands upon Panait and Luke's solution for transitioning into a new state ^[9, eq. 2], where instead of strictly following the greatest utility values, an agent will take the peripheral pheromone type, q , into account too. With this equation, the desirability of a given s'' can be reduced by how much or how little the utility value of the opposing pheromone is, $U_q(s'')$, according to a popularity factor α . This then introduces logic where agents will ignore pheromone trails that are highly popular, and instead explore out either in search of other food sources, or follow a different pheromone trail. A state is then chosen to transition to by selecting a w with the greatest value in W , through Equation 6. In the case where multiple weights are equal, a random selection of those maximum values is performed.

The popularity factor, α , is a constant value greater than 0, and defines how many agents must be populating a given pheromone trail before it is ignored. The two graphs in Figure 3 demonstrate how the differing parameters of Equation 5 can effect the resulting weights. Looking at Figure 3b, this graph shows the resulting weights when $U_p(s'') = 1$, given varying values of α . It can be seen that as α increases, the plateau at the top of each line grows longer, which is the primary segment of this equation that defines how populated a pheromone trail must be for an agent to ignore it. This behaviour is caused by the fact that with a greater value of α , the value of w drops off and reduces much later as the peripheral pheromone utility value increases.

Figure 3a then shows how the resulting weights are ef-

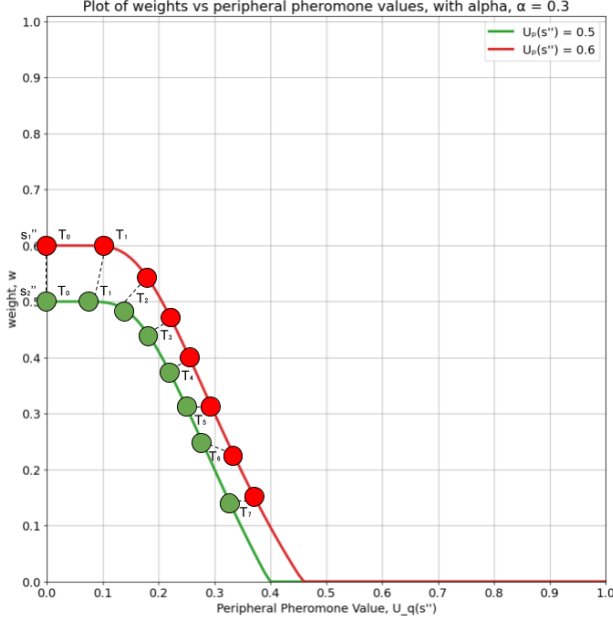


Figure 4: A timeline graph showcasing how two states, s'_1 and s'_2 , that are either side of a goal state, differ in resulting weights as time T progresses and more agents visit the goal state. As time progresses, the difference in weight between the two states reduces, gradually leading to s'_2 becoming more favourable at T_6 . After this point, s'_1 and s'_2 continuously alternate in being the most favourable, leading to a balanced pair of state transitions.
All variables are arbitrary in this figure

affected by differing values of the focal pheromone utility value, given a constant popularity factor of $\alpha = 0.3$. As expected, for each value of $U_p(s'')$, the curves follow similar trajectories, and reach a weight of zero sooner the smaller the value of $U_p(s'')$. However, it is important to notice that no matter the size of the focal utility value, the plateau at the top of each line starts to drop off at the same utility value of the peripheral pheromone, $U_q(s'')$. This constant drop off threshold helps towards allowing the agents to self-balance their pheromone trails, ensuring that once the popularity threshold has been achieved, the trail will always maintain that popularity.

For example, take two state transitions on opposing sides of s' , s'_1 and s'_2 . At T_0 , the agent has arrived at s' from the direction of s'_1 (also known as s in fig. 2) while depositing p_{nest} , and is at their goal state, which is a newly discovered food node. The agent has now switched internal states to follow back from where it came. As this food node is newly discovered, there are no surrounding pheromones that guide agents towards it, meaning the peripheral pheromone utility values, $\{U_q(s'') \mid s'' \in S''\}$, in the surrounding area all equal to 0. We also know that due to pheromone dispersion (eq. 2), state s'_2 will have some $U_{p_{nest}}$ value. From this, we can infer that:

- Focal Pheromone: p_{nest}
- Peripheral Pheromone: p_{food}
- $U_{p_{nest}}(s'_1) > U_{p_{nest}}(s'_2)$

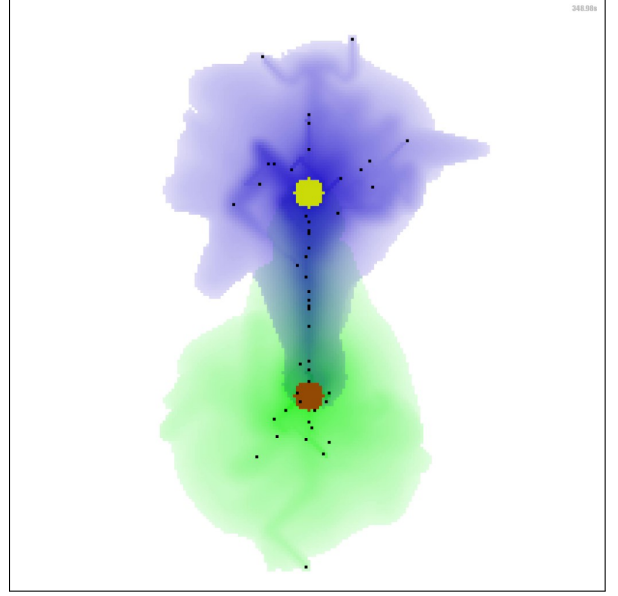


Figure 5: Environment visualisation of a duel-node setup for the colony of agents to develop a pheromone trail between the two nodes. The brown top circle represents a nest node, with the yellow bottom circle being the food node. The green and blue colours (densely surrounding the nest and food nodes respectively) represent utility values of the two different types of pheromones, p_{nest} and p_{food} respectively. Utility values are amplified through $U_{p,scaled} = U_p^{0.1}$ only for the environment visualisation. The actual values used in the simulation remain unaffected.

$$- U_{p_{food}}(s'_1) = U_{p_{food}}(s'_2) = 0$$

We can now use Figure 4 to easily find the resulting weights when plugging these parameters into Equation 5 for s'_1 and s'_2 , looking at T_0 . It is found that $w(s'_1) = U_{p_{nest}}(s'_1)$ and $w(s'_2) = U_{p_{nest}}(s'_2)$, due to the fact that $U_{p_{food}}(s'_1)$ and $U_{p_{food}}(s'_2)$ both equal 0, meaning nothing is taken away from our two $U_{p_{nest}}$ values in Equation 5. So then as we know that $U_{p_{nest}}(s'_1) > U_{p_{nest}}(s'_2)$, our agent will transition into s'_1 and follow the path that it made on the way to this food node back towards the nest, while depositing p_{food} in parallel.

Once the agent returns to the nest node, a pheromone trail has been created using p_{food} that connects the nest and food nodes, with a gradient that ascends towards the food node. This allows other agents to detect and follow this pheromone trail for reinforcement, leading to the $U_{p_{nest}}$ values surrounding the food node, in addition to $U_{p_{food}}$ values, to gradually grow. It is important to note though, that with the combination of the diffusion and evaporation effects on pheromones (eqs. 2 and 3), as well as the decay rate, γ , when depositing pheromones (eq. 4), these utility values for p_{food} remain to be very small values, approximately in the range of $[0.3, 0.5]$, when only a few states away from the food node. All figures in this paper showcasing this model have the visual representation of pheromones scaled up using $U_{p,scaled} = U_p^{0.1}$, in order to make them clearer in the visual environment. Without

this scaling up, the majority of utility values will be too small to see.

So, now that the two pheromone utilities are gradually increasing around the food source, the colonies self-balancing of the pheromone trail begins. With the first few arrivals of agents to the newly found food node (T_1 on Figure 4), the value of $w(s'_1)$ remains to be greater than $w(s''_2)$. Only now, the values of $U_{p_{food}}(s'_1)$ and $U_{p_{food}}(s''_2)$ are rising, causing Equation 5 to have more of an effect on $U_{p_{nest}}$ values.

As time T progresses, $w(s'_1)$ continues to be greater than $w(s''_2)$, but the difference between the two gradually decreases as $U_{p_{food}}$ at s'_1 increases faster than at s''_2 , due to all agents transitioning and depositing pheromones into s'_1 . An equilibrium is eventually hit at T_5 , leading to half of the agents transitioning into s''_2 instead. At T_6 , the population threshold has been exceeded and $w(s''_2)$ now has a greater value, leading to agents favouring the transition into state s''_2 instead. This causes a sudden increase in $U_{p_{food}}(s''_2)$, thus pushing some agents back towards s'_1 . At T_7 , self-balancing of this trail has been achieved by the colony, with the two values of $w(s'_1)$ and $w(s''_2)$ alternating in which one is marginally greater than the other.

The resulting environment of this outlined scenario can be seen in Figure 5. Here we can identify that a strong, stable pheromone trail guiding agents to and from the nest and food nodes has been developed, with p_{food} ascending in concentration towards the food node, and p_{nest} ascending towards the nest node. The path between the two nodes is noticeably optimized into the most efficient route, a direct straight line between the two, indicated by the black dots that represent individual agents forming a uniform line in the middle. We can also see that, due to the trail connecting the two nodes being populated enough to keep it well maintained, spare agents are exploring in opposing directions at both the nest node and food node, in search of additional food nodes. In the event that too many agents continue to forage instead of following the trail back the way they came, agents returning from foraging after running out of pheromones ($U_p(s')$ from eq. 4 equals 0), or getting lost, will detect the slack and take over the maintenance.

This complex equation is needed to ensure that individual agents can determine whether they should return to their start state after reaching their goal state, or to continue foraging, while remaining within the bounds of indirect communication with other agents through the use of pheromones.

Once extended foraging begins from a given food node, that node can be treated exactly the same as a nest node. Agents will use it as another “hub”, where they can disperse in search of a new food node just like they would at the beginning of the simulation at the nest node. At this point, it is easier to think of the two pheromone types as p_A and p_B , replacing p_{nest} and p_{food} respectively. Referring to Figure 5, this would mean that the green pheromone is p_A , and the blue is p_B .

When dispersing from the recently discovered food node, $N_{food,1}$, in search of additional food nodes, agents will be depositing p_B , as that is the pheromone that has been “as-

signed” to that food node. Once an agent discovers a new food node, $N_{food,2}$, their internal state will switch as normal, leading them to now be following the p_B pheromone they laid on their way to this node, and depositing the p_A pheromone, which is now the “assigned” pheromone type for this new food node. Upon arrival back to the previous food node, $N_{food,1}$, other agents will detect this new, low-populated trail and start traversing it to reinforce it. From here, the exact same process as to what was explained with a 2-node system is carried out.

With this, the model is capable of creating a chain of interconnected nodes. Assuming the colony has a high enough agent population, there is theoretically no limit as to how many nodes can be added to the environment for agents to discover and populate.

4 Results

Using these new update and transition rules (eqs. 4 to 6), this model is capable of allowing a colony of agents with one source nest to forage for multiple food nodes simultaneously, not only creating pheromone trails from nest to food nodes, but also food to food nodes. Although this was possible through Panait and Luke’s model, their model required a unique pheromone type for every node in the environment, leading to linearly scaled computational usage for every new node. This becomes an issue on large scale environments with many individual nodes, due to the higher demand for computational power. In addition to their model, Hartman et al.^[15] and Malíčková et al.^[16] both proposed alternative methods for a multi-node environment while utilizing a constant amount of pheromones, but both relied on the fact that all food nodes were equidistant from the nest node, and that no connection could be made from food to food nodes.

Parameter	Value	Description
$W \times H$	200×200	Width and height of the grid world
dt	0.01	Time step of every simulation update
A	$25AN^{-1}$	Amount of agents in a given colony, per node in the environment (AN^{-1})
D	0.7	Pheromone diffusion coefficient
k	0.1	Pheromone evaporation coefficient
γ	0.9	Pheromone deposit decay rate
α	0.3	Pheromone trail following popularity factor

Table 1: *Parameter values used in all simulations*

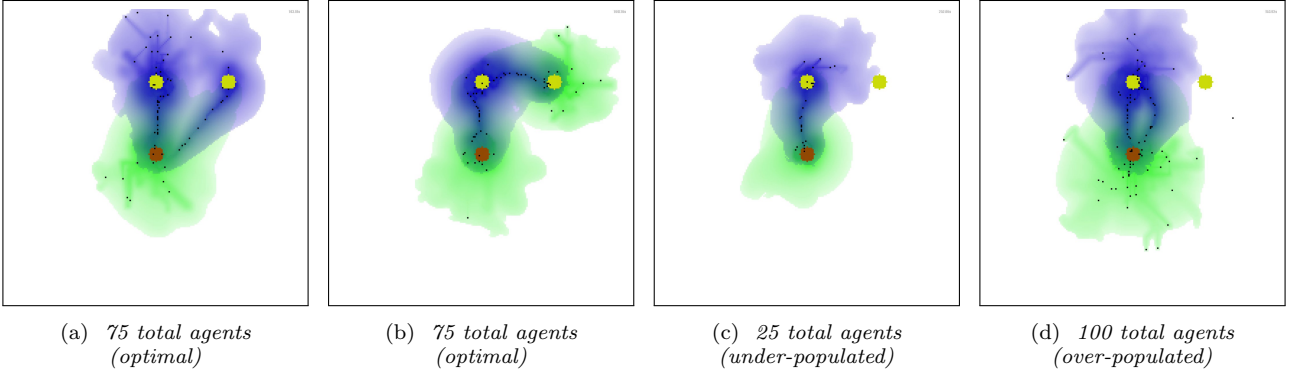


Figure 6: Each of these four figures are the resulting pheromone trails in an environment with one nest node and two food nodes with positions that represent the three vertices of a right-angled triangle. The figures are to demonstrate the impact of over- or under-populating the colony in simulations of this model. Both (a) and (b) have populations of 75 agents each, which is the optimal colony population for a three-node environment, according to Table 1. (c) is an example of under-population, at 25 agents, leading to the inability to forage for the second food node, and (d) is an example of over-population with 100 agents, leading to multiple pheromone trails being formed to connect the same two nodes.

The model outlined in this paper avoids the computational limitations by only utilizing two pheromone types throughout, no matter how many nodes are placed into the environment. This allows for a continuous requirement of computational power for any simulation setup, as well as a simpler agent model, as their decision making does not need to factor in and keep track of many different internal states, instead only requiring two states. The equidistant nodes limitation is also avoided through the use of an alternative equation for following pheromones, and can successfully form pheromone trails between any number of nodes with any distance separating them (assuming it is within range of the agents' reach).

4.1 Parameters

Parameters for all simulations run on this paper's model are outlined in Table 1. The environment is set up to have a width and height of 200×200 , with both pheromone types having diffusion (D in eq. 2) and evaporation (k in eq. 3) factors of 0.7 and 0.1 respectively. Agents will decay the pheromone that they are depositing at a rate of γ , which has the value of 0.9, and will follow a given pheromone trail in respect to the popularity factor, α , at a value of 0.3. An optimal number of agents, A , was found to be 25 agents per node (AN^{-1}).

Figure 6 shows examples that present the resulting pheromone trails of four different simulations in an environment with one nest node and two food nodes, laid out in the shape of a right-angle triangle. Each of the four examples were simulated with varying amounts of colony populations, where Figures 6a and 6b have 75 total agents each, 6c has 25 agents, and 6d has a population of 100. It can be seen in these figures that populations below $25AN^{-1}$ (fig. 6c) would lead to the colony of agents unable to explore much beyond a given node due to the fact that the majority of agents are needed to maintain the trail leading to that node. Populations above $25AN^{-1}$ (fig. 6d) would cause over-reinforcement of trails, leading to mul-

tle pheromone paths being created between two nodes. With a population of $25AN^{-1}$ (figs. 6a and 6b), the colony has the perfect population count in order to be able to continue a search for food nodes beyond what has already been discovered, while also ensuring that there are not too many agents that lead to over-maintaining pheromone trails.

4.2 Network formation

Simulation tests of the model have proven to show that a colony of agents are capable of creating a network of pheromone connections between multiple nodes, all while only utilizing two pheromone types, p_A and p_B . The observed network shapes tend to favour a low cost (total length of edges) structure in sacrifice of it's robustness (resilience to the random removal of edges). These results differ to Panait and Luke's solution for multi-node environments, where their model resulted in network structures with high cost and high robustness.

In Figure 7, an environment similar to Figure 6 is presented, with three total nodes in the shape of a right-angled triangle. Each image, figs. 7a to 7e, provide a chronological step through of each milestone the colony achieves in creating a network connecting all the nodes in the environment. The simulation starts at Figure 7a, with all the agents exiting the nest node in uniformly random directions, each depositing a pheromone that guides back towards the nest node, p_A . The agents are also trying to follow p_B , but as this is the start of the simulation, p_B does not exist yet.

At Figure 7b, a food node has been discovered by an agent. This agent switches it's internal state, such that it is now depositing p_B and following p_A , and travels back towards the nest while creating a trail that guides other agents towards the new food source it just discovered. These other agents that join onto this pheromone trail help reinforce and maintain the path, all while gradually

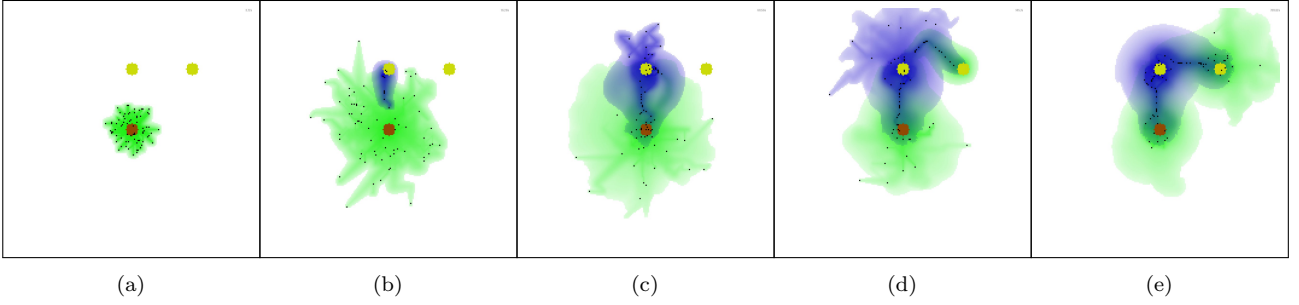


Figure 7: *The foraging sequence of a 3-node environment with each node set up in the shape of a right-angled triangle. The brown circle represents the nest node, with the food nodes shown as the two yellow circles. (a) Agents leave the nest while depositing pheromone p_A (green) to lead back to the nest. (b) Agents discover a food source and begin their return back to the nest node, following p_A back to the nest while depositing pheromone p_B (blue) leading towards that food node. (c) A stable trail is established connecting the newly found food node to the nest node. (d) A second food node is discovered. (e) That second trail is optimized.*

optimizing it, which is then achieved in Figure 7c.

Agents are now able to continue the foraging search for additional food nodes at this point in the simulation, and can be seen expanding outwards from both the nest node and discovered food node. This extended search branching off of nodes leads to the discovery of the second food node to the right of the first that the colony discovered, as seen in Figure 7d. This new pheromone path is strong and stable, however it also has an arching shape that is clearly not optimal. Over time, the agents gradually bring the curve of that arching shape down to a more level path, resulting in the optimized trail observed in Figure 7e.

The final network shape presented in Figure 7 is well maintained by the agents, with strong and efficient node connections. However, this is not the only network shape the colony can form for this given environment setup. Figures 6a and 6b demonstrate how the colony of agents can create a network of pheromone trails with varying shapes. Although it is not the focus of this paper, given an environment where the transportation of food back to the nest is a required factor, and that food nodes have a limited amount of food, these two simulation outcomes differ in their pros and cons. Figure 6a showcases a solution that provides a high inflow of food into the nest, depleting resources across all the food nodes in a shorter period of time. Whereas Figure 6b demonstrates a low inflow of food into the nest (roughly half that of fig. 6a), but depletes resources from all food nodes across a longer period of time. Both resulting networks are viable for this model, and demonstrates that there is no one “right” answer for these simulations. Each new simulation may produce varying results, capable of exploring all *efficient* network shapes given enough simulation iterations.

All the examples and figures in this paper so far have showcased environments with only two or three total nodes in the environment, but it is very easy to expand upon that. No agent needs to be changed in any way. Their total internal states remain to be only two, as the total unique pheromones utilized in the simulation does not increase from two. It is as simple as adding additional food nodes into the environment, and running the simulation.

Additional node simulations can be seen in Figure 8,

where an alternative layout of the three-node environment is presented, as well as a four-node and five-node environment setup. These node layouts are the same as what can be found in Panait and Luke’s experiments for their solution to exploring multiple nodes in an environment [9, fig. 6].

With this model’s solution to the three-node tour (fig. 8a), we can see that the two food nodes are only connected to the nest node, and not also connected to each other. This is an example of a low cost, low robustness network shape, as the total length of edges is low, meaning less resources are required to construct the network, but the network is then prone to breaking down, as if any of the two edges are broken, the node that edge leads to is no longer accessible. Whereas Panait and Luke’s solution is an example of high cost and high robustness, due to those two food nodes also being connected to one another. Their resulting network shape can withstand a broken edge without losing access to any node, however it requires a higher amount of resources to construct.

For the four-node tour shown in Figure 8b, the solution presented is exactly the same as Panait and Luke’s model. All four corners are connected to their neighbours, forming a square shaped network. This solution is a good example of the cost and robustness rating for the network being as balanced as it can be, as the network can afford to lose one connecting edge while remaining to not be the most expensive shape for this given layout. Each node could connect diagonally to the opposing node for greater robustness, but depending on the application, that may be an unnecessarily high cost.

Figure 8c presents a solution for a five-node tour. This network structure has the lowest possible cost for this node layout, and thus the lowest robustness. If, for example, the right-hand edge from the nest node breaks, those three food nodes that the broken edge leads to will no longer be accessible. However, Panait and Luke’s solution to the five-node tour provides a more robust structure, for a higher cost. In their network shape, several nodes are interconnected with one another, accounting for the possibility that an edge could be broken without isolating any nodes.

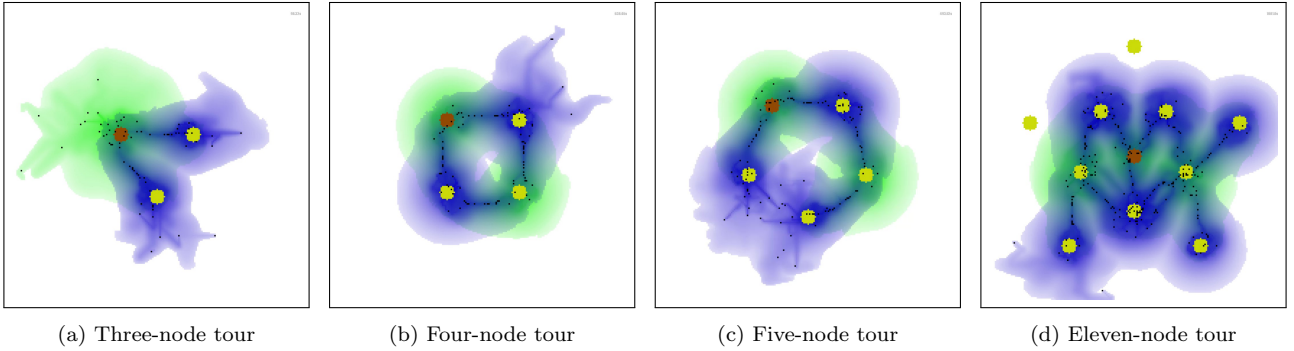


Figure 8: *Colony tours with three, four, five, and eleven nodes in the environment. (a,b,c) present the solutions found by the model outlined in this paper for node layouts used in Panait and Luke’s testing of their “multiple waypoints” model. (d) demonstrates the capabilities of the model on a larger scale of nodes placed in the environment, extending the experiments performed by Panait and Luke.*

Figure 8 also showcases an example of an eleven-node environment, where ten food nodes surround a central nest node (fig. 8d). It can be seen that the majority of nodes are robustly connected, providing multiple paths that lead to each node. However, there are also two food nodes, the far left and top-most nodes, that have not been found by the agents. After leaving the simulation to run for 17 simulation minutes, a small amount of agents came close to discovering these two nodes, but never successfully found them. This could be due to a variety of factors that work together to prevent agents from further discovering nodes.

Due to there existing five food nodes in the “inner circle” around the nest node, a high volume of agents get absorbed into multiple pheromone trails connecting those five food nodes. It is observed in the inner bottom-right corner, two trails have been formed to connect just two food nodes together, where the agents then decide that they need to maintain both paths, which should not be the case. This double trail leads to twice the amount of agents being utilized to maintain a connection between these two nodes, causing there to be less agents available to explore beyond nodes and discover additional food nodes.

This issue could be tackled with a higher colony population. Yes, this would initially lead to a higher quantity of double-paths between two nodes on the inner circle of food nodes, but as time goes on these double-paths should be optimized into single-paths. As the paths get optimized, more agents will gradually become available to explore outwards, leading to a greater chance of these outer food nodes being discovered in full. Another way to solve this problem is to adjust the popularity factor, α . This would provide the same effect as increasing or decreasing the colony population, but may also lead to more prominent and consistent double-paths between nodes in the early stages of the simulation.

In lightly testing these hypothesised parameter adjustments to successfully create a network structure connecting an eleven-node layout (as seen in fig. 8d), it was found that the adjustments did not assist much. The resulting network shapes after 15 simulation minutes remained to exclude two or three of the outer food nodes. It is hypothesised that there exists too many agents in the early stages

of the simulation, leading to the colony forming multiple pheromone trails between two given inner-nodes. These unnecessary trails proceed to experience constant maintenance, with some paths merging into one briefly before splitting up again. This allows for some outer-nodes to be discovered, but once only a few of them are discovered, the pheromone trails in the network, including the unnecessary trails, result in being well balanced, meaning very few agents can explore outwards in search of more food nodes.

5 Conclusions and Future Work

The use of indirect communication between agents through pheromone depositing and following provides a simple method for collective communication and coordination on a large scale, and is an elegant solution to multi-agent engineering contexts. This study introduces an alternative foraging approach in a multi-node environment, in order for agents (ants) to form a network of pheromone trails that connect all nodes together. This alternative approach focuses on ensuring the model only requires two unique pheromone types, as well as being capable of not only creating nest-to-food pheromone trails, but also food-to-food trails, allowing for a simpler and more computationally efficient model. Akin to pre-existing models^[9,15], agent locations are viewed as external states, and the utilities of those states representing pheromone concentrations, similar to reinforcement learning.

5.1 Model limitations

With addition of conventional dispersion and evaporation behaviours of pheromones, the model was able to achieve the task of forming pheromone trails that connected all nodes in the environment on a small scale (two to five total nodes). However, when the colony of agents was placed into a large scale environment consisting of eleven total nodes, they were unable to complete the network, only connecting together the majority of nodes and leaving out a small portion of two to three nodes. A potential solution to this issue could be to release agents from the

nest in batches, where a new batch of agents is released (in addition to those already foraging) every time a new node is discovered. This could reduce the problem outlined in Section 4.2 for an eleven-node environment, where too many agents within the inner circle of nodes led to multiple pheromone trails being formed between two given nodes.

Due to the logic behind agents following a given pheromone trail is determined by its popularity, this leads to the possibility of under- and over-population of the colony. Under-populated colonies will not be able to discover and retain knowledge of multiple food nodes due to there being an insufficient amount of agents to maintain all the paths. Whereas an over-populated colony will construct additional unnecessary pheromone trails between nodes, leading to agents maintaining a path that it not required and thus less agents available for extended foraging for new food nodes. Similar undesirable outcomes of under- and over-populated colonies can also be caused by improperly tuning the popularity factor, α .

Although this model has performed well compared to alternative methods, it has not been experimented on in more complex environments. These complex environments would include obstructions blocking the path of agents, and moving food nodes (as tested by Panait and Luke^[9]), as well as possible limited resources at food nodes. It is hypothesised that the addition of obstructions to the environment would cause agents to produce two pheromone trails that traverse around either side of the obstruction in order to reach the target node (given that both routes are within range of the previous node), forming an inefficient connection between the two nodes. In addition to this, it is believed that there will be inefficiencies with a moving food node, as the agents can be very slow with path optimizing. However, with depleting food nodes, the agents should easily be able to detect an empty food node and begin to ignore the location, allowing the pheromones around the area to gradually fade away.

Despite the minor issues, this unique model has performed well in achieving its baseline goals. The capability of agents forming multiple pheromone trails between multiple nodes with only using two pheromone types closely relates to behaviours found in nature^[13,14], bringing forth a more accurate model for simulating ants.

5.2 Future Work

So far, this model has only experienced light-weight experiments in order to ensure functionality for the core focus of the model – to generate a network of pheromone trails between any number of nodes. The first extension to this model would be to fix the struggle with connecting all nodes in the environment on a large scale, making use of batch releasing of agents. Once full functionality has been achieved, experiments with obstructions placed into the environment will take place. Given that the hypothesis outlined previously about how the colony will behave when manoeuvring around an obstruction is correct, methods as to how to avoid the inefficient path finding will be explored to correct this behaviour.

Once these two corrections have been implemented into

the model, it has potential to achieve some greatly practical applications. For example, the model could be put to use in designing an optimized city transport network, where each node would represent a station for the desired transportation vehicle to stop at. The model parameters would be tuned just right for the desired cost and robustness of the network, as well as obstacles being placed in areas where the network cannot go, and from that information the colony of agents will forage and create an efficient transport network to be used in the real world.

Beyond that, the idea of adaptive parameter tuning^[19] can be explored, where agents will have the capability to dynamically adjust parameters such as the popularity factor α , and pheromone depositing decay rate γ . This will expand even further upon the model's ability to efficiently and effectively generate networks connecting all nodes, possibly leading to the ability of only requiring a constant population of agents, despite how many nodes are placed into the environment, instead of the current 25 agents per node (table 1). This improvement would lead to the the entire model having a constant computational complexity, proving to be the most efficient solution to a multi-node environment problem.

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